Anomaly Detection in Ship Engine Data

Table of Contents

1. Introduction ***1***

1.1 Approach **2**

2. Exploratory Data Analysis ***2***

3. Methods ***4***

4. Key Findings and Results ***4***

5. Interpretation of Anomaly Detection Results ***5***

6. Evaluation of 2D PCA Plot Visualisations ***6***

..... ***7***

..... ***7***

7. Conclusion ***7***

8. Recommendations ***7***

References ***8***

1. Introduction

This project addresses a real-world anomaly detection challenge in the maritime industry. A dataset of 19,535 continuous sensor readings from a ship's engine was provided for analysis, spanning six key metrics: Engine RPM, Lubrication Oil Pressure, Fuel Pressure, Coolant Pressure, Lubrication Oil Temperature, and Coolant Temperature.

The goal of this project is to develop a robust, automated anomaly detection system capable of identifying potentially problematic engine states before they escalate, enabling the business to shift from reactive to predictive maintenance.

Anomalies are expected to represent approximately 1–5% of all readings. A critical business consideration is that individual sensor extremes may not indicate a problem in isolation. For example, high Engine RPM during acceleration may be entirely normal, but combined with high coolant temperature, it becomes an issue. Therefore, anomalies are most meaningful when multiple sensors deviate simultaneously, reflecting a genuine fault pattern rather than a routine operational fluctuation.

1.1 Approach

Three methods were applied in sequence, progressing from statistical to machine learning approaches, as illustrated below:

Stage 1 Data Loading & EDA	**Stage 2** IQR Statistical Detection	**Stage 3** ML Detection (SVM & IF)

Data was first explored through Exploratory Data Analysis (EDA) to understand distributions, central tendency, and feature relationships. The Interquartile Range (IQR) method was then applied as a statistical baseline. Finally, two machine learning models (One-Class SVM and Isolation Forest) were trained across all six features. A cross-method comparison then identified the highest-confidence anomalies by finding agreement between all three methods.

IQR was chosen as a statistical baseline because it requires no distributional assumptions and is easily interpretable. One-Class SVM was selected for its ability to define a tight boundary around the normal operating region. Isolation Forest was included because of its computational efficiency and robustness to high-dimensional data.

2. Exploratory Data Analysis

Initial data quality checks confirmed that the dataset is clean. There are no missing values and no duplicate records across all 19,535 observations. All six features are continuous numerical variables, confirming suitability for direct statistical and machine learning analysis.

Exploratory Data Analysis revealed that all six features are positively skewed, and the mean exceeds the median in every case. This is seen in Figure 1. Engine RPM shows the most pronounced skew (skewness = 0.935, mean-median gap = 45 points). Coolant Temperature contains a maximum value of 195°C; nearly 120°C above its mean of 78°C, demonstrating that descriptive statistics alone can mask extreme individual values in large datasets.

Boxplot analysis in Figure 2 confirmed the presence of extreme outliers across all features, consistent with the skewed distributions observed in the histograms. Engine RPM and Coolant Temperature showed the most pronounced outlier spread, supporting the case for a multi-feature detection approach rather than single-sensor thresholding.

The correlation heatmap in Figure 3 confirmed no significant relationships between any pair of features, establishing that multivariate detection is essential. Single-feature analysis would miss anomalies that arise only when combinations of features behave unusually

together. The absence of correlation means that anomalies emerging from combinations of features will be invisible to the IQR method, motivating the use of machine learning models.

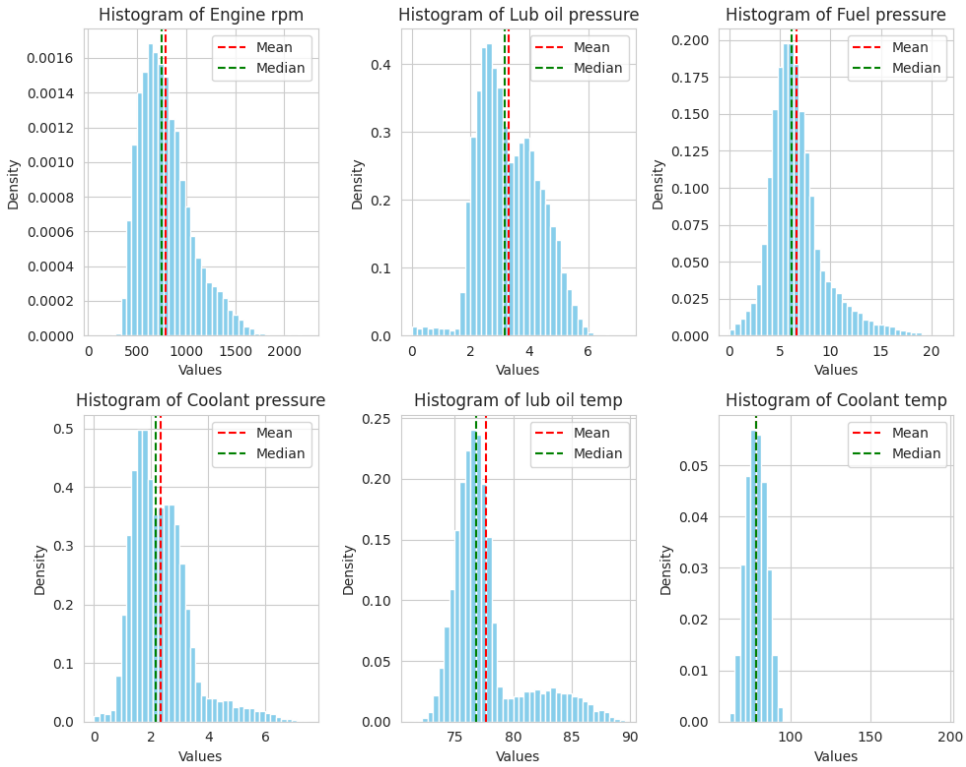

Figure 1: Histogram plot of all features

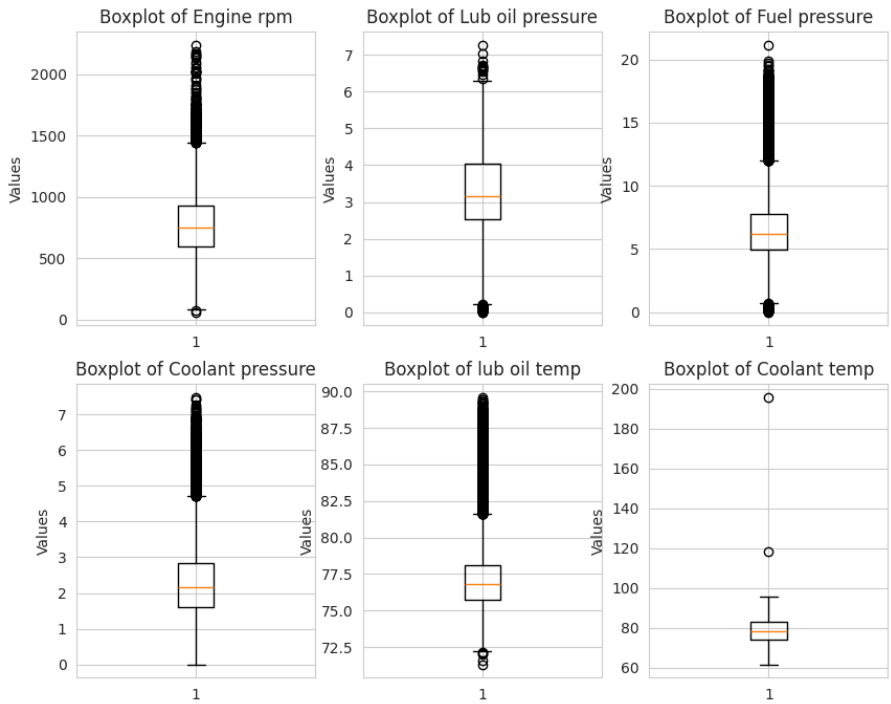

Figure 2: Boxplots of all the features value distributions

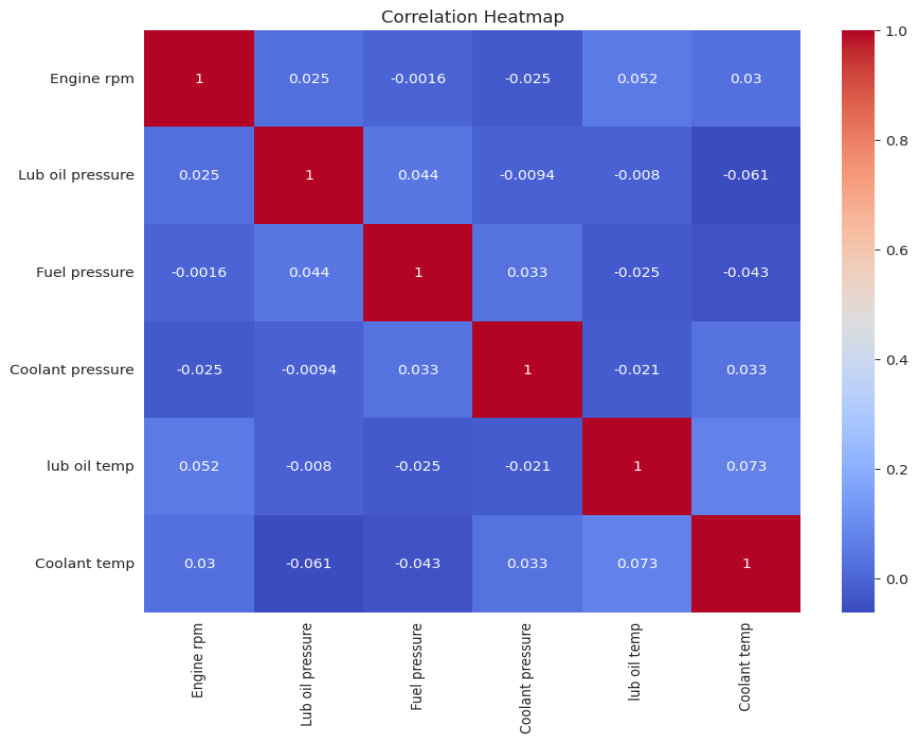

Figure 3: Correlation Matrix of all features

3. Methods

For IQR, a multi-feature threshold was applied. Rather than flagging single-feature outliers, data points were only classified as anomalous if two or more features simultaneously exceeded their IQR bounds. At T=1, 23.73% of the data is flagged, too high for the set parameter. T=2 yields 2.16%, within the expected range. T=3 collapses to just 0.06%, which is too conservative for our parameter.

For the machine learning models, data was first standardised using StandardScaler, due to One Class SVM being sensitive to feature magnitude. Principal Component Analysis (PCA) was then applied separately for visualisation purposes only, reducing the six-dimensional feature space to two components. Both models were configured conservatively: One-Class SVM with nu=0.01 and Isolation Forest with contamination=0.01, representing the lowest parameter values yielding an anomaly rate within the 1–5% target. This minimises false positives, a key concern in a production engineering context where each false alarm consumes valuable engineer time.

4. Key Findings and Results

Table 1 compares the anomaly counts produced by each detection method:

Table 1: Anomaly Detection Results by Method

Detection Method	Anomalies Flagged	% of Dataset
IQR (Threshold T=2)	422	2.16%
One-Class SVM (nu=0.01)	198	1.01%
Isolation Forest (contamination=0.01)	196	1.00%
Two-Method Agreement	176	0.90%
All-Method Agreement	58	0.30%

Table 2 shows the mean feature values for Isolation Forest-flagged anomalies versus normal readings, revealing the primary drivers of anomalous engine states:

Table 2: Anomaly Profile — Normal vs Anomalous Engine Readings

Feature	Normal Mean	Anomaly Mean	% Change
Engine RPM	788.5	1,058.1	+34.2%
Lubrication Oil Pressure	3.302	3.449	+4.4%
Fuel Pressure	6.617	10.513	+58.9%
Coolant Pressure	2.323	3.602	+55.1%
Lub Oil Temperature	77.6°C	82.1°C	+5.8%
Coolant Temperature	78.4°C	79.2°C	+0.9%

Fuel Pressure and Coolant Pressure show the largest deviations of +58.9% and +55.1%, respectively. Lubrication Oil Pressure and Coolant Temperature show minimal mean differences. Engine RPM is elevated by 34.2% in anomalous states. This pattern indicates that anomalies are not driven by a single runaway sensor, but by a cluster of indicators rising together.

5. Interpretation of Anomaly Detection Results

The close agreement between One-Class SVM (198 anomalies) and Isolation Forest (196 anomalies) provides strong cross-validation. Despite using fundamentally different algorithms, SVM defines a hypersphere boundary around the normal region, while Isolation Forest isolates anomalous points through random partitioning; both models converge on nearly identical conclusions. This convergence substantially increases confidence in the flagged cases.

The 58 anomalies flagged by all three methods represent the highest-confidence cases in the dataset. The anomaly profile in Table 2 reveals a consistent pattern: anomalous engine states are characterised by elevated Fuel Pressure, Coolant Pressure, and Engine RPM

occurring simultaneously. This is the multi-feature interaction that the business context identified as a genuine operational risk.

The divergence between IQR (422 anomalies) and the ML methods (196-198) is expected. IQR applies fixed univariate bounds per feature and cannot capture anomalies that only emerge from feature combinations. The 176 two-method agreement anomalies form a useful mid-confidence tier, primarily representing cases where SVM and Isolation Forest agree but IQR does not, suggesting the anomaly is subtle enough to evade univariate detection.

6. Evaluation of 2D PCA Plot Visualisations

The 2D PCA plots seen in Figure 3 and Figure 4 reduce the six-dimensional feature space to two principal components for visualisation. However, these two components capture only 36.7% of the total variance (PC1: 19.0%, PC2: 17.7%), meaning 63% of the data's structural information is discarded in the process.

Consequently, the plots have significant limitations. Anomalies detected in the full six-dimensional space may not appear visually separated from normal data in 2D if the feature combinations driving their anomalous behaviour are captured in the discarded components. The plots are most effective at confirming the most extreme outliers and provide a useful high-level illustration of model behaviour. However, they should not be used as a standalone detection tool. For accurate anomaly detection, the full-dimensional models remain the authoritative source.

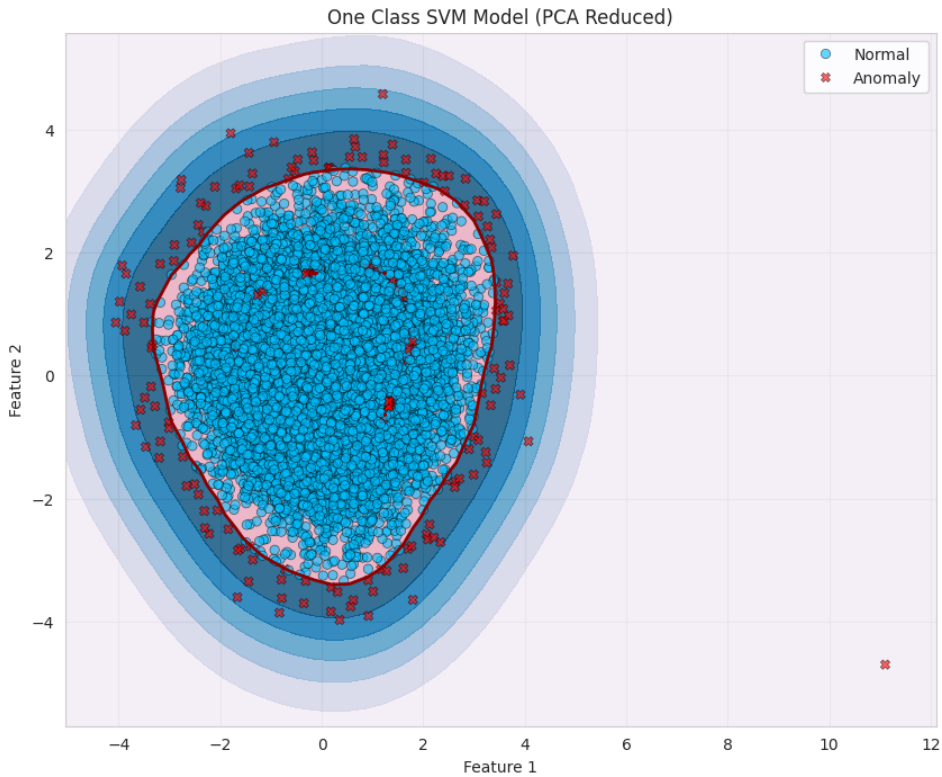

Figure 3: One-Class SVM with PCA

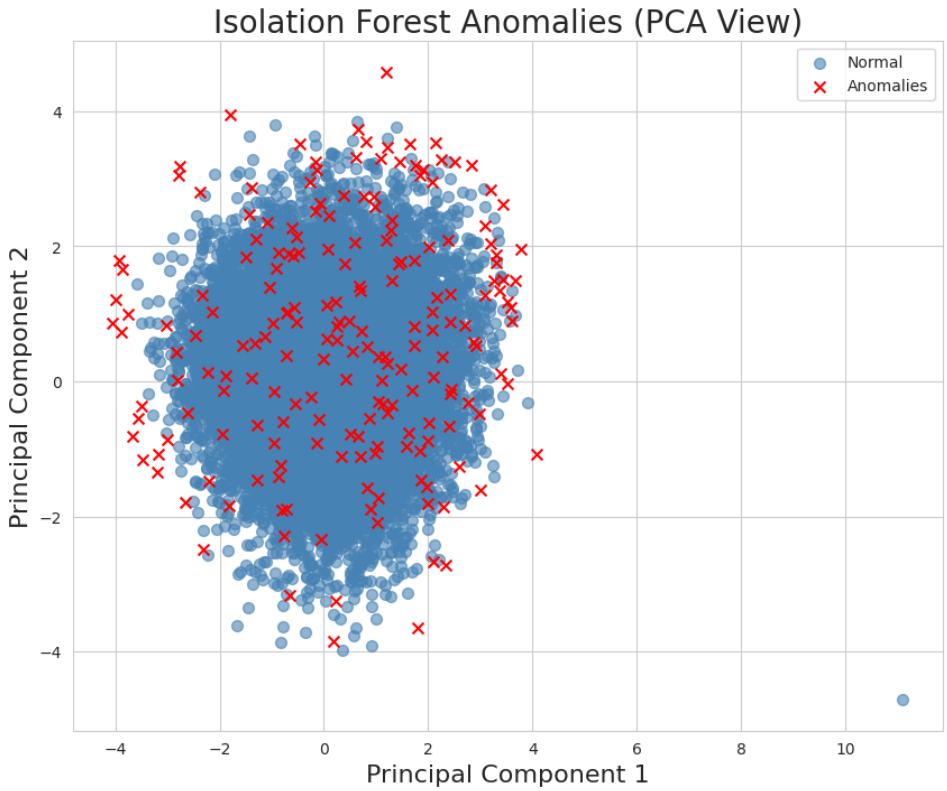

Figure 4: Isolation Forest with PCA

7. Conclusion

This analysis successfully developed a multi-method anomaly detection system for ship engine monitoring. Across three independent detection approaches (IQR, One-Class SVM, and Isolation Forest) 58 high-confidence anomalies were identified by unanimous agreement, characterised by simultaneous elevation of Fuel Pressure (+58.9%), Coolant Pressure (+55.1%), and Engine RPM (+34.2%). The close agreement between One-Class SVM (198 anomalies) and Isolation Forest (196 anomalies) provides strong cross-validation of the findings, confirming that these are genuine multi-feature fault patterns rather than statistical noise. Isolation Forest is recommended as the primary production model given its computational efficiency and independently verified results. Deploying this system would allow the business to shift from reactive to predictive maintenance.

8. Recommendations

Based on the analysis, the following recommendations are made to the engineering and operations teams:

1. **Immediate investigation:** Prioritise the 58 consensus anomalies flagged by all three methods for immediate engineering review. The probability of false positives in this group is very low, given the tri-method agreement.
2. **Scheduled inspection:** Log the 176 two-method agreement anomalies for inspection during the next planned maintenance window.

3. **Primary monitoring indicators:** Designate Fuel Pressure and Coolant Pressure as the primary real-time alert triggers, given their +58.9% and +55.1% deviation in anomalous states.
4. **Production deployment:** Deploy Isolation Forest as the primary production model due to its computational efficiency, scalability to large streaming datasets, and results independently confirmed by One-Class SVM.
5. **Domain expert review:** IQR-only anomalies, those not confirmed by either ML model, should be reviewed by ship engineers to determine whether they represent genuine operational concerns or are just statistical outliers caused by the IQR method's univariate approach.

References

Devabrat (2022). Ship Engine Sensor Dataset. GitHub. Available at: https://github.com/fourthrevlxd/cam_dsb